

NO.	SOLUTION	MARKS	REMARKS
PART A			
1	(a) $\lim_{x \rightarrow 3} \sqrt[3]{\frac{x^2 - 1}{2x^3 + x^2 + 1}} = \sqrt[3]{\lim_{x \rightarrow 3} \frac{x^2 - 1}{2x^3 + x^2 + 1}}$ $= \sqrt[3]{\frac{3^2 - 1}{2(3)^3 + (3)^2 + 1}}$ $= \sqrt[3]{\frac{8}{64}}$ $= \frac{1}{2}$ (b) $\lim_{x \rightarrow 0} \frac{3x^2 + mx}{4x - 4x^2} = 2$ $\lim_{x \rightarrow 0} \frac{x(3x + m)}{4x(1 - x)} = 2$ $\frac{3(0) + m}{4(1 - 0)} = 2$ $m = 8$	K1 K1 J1 K1 K1 J1	Apply properties Substitute value factorise

2

NO.	SOLUTION	MARKS	REMARKS
	$\frac{dy}{dx} = \left(x^{\frac{1}{2}}\right) \left(2^{\sin x} \cos x \ln 2\right) + \left(2^{\sin x}\right) \left(\frac{1}{2} x^{-\frac{1}{2}}\right)$ $\frac{dy}{dx} = \left(x^{-\frac{1}{2}} 2^{\sin x}\right) \left(x \cos x \ln 2 + \frac{1}{2}\right)$	K1 J1	
	(b) $y = \ln \left(x \sqrt{1+x^2}\right)$ $y = \ln x + \ln \sqrt{1+x^2}$ $y = \ln x + \ln \left(1+x^2\right)^{\frac{1}{2}}$ $y = \ln x + \frac{1}{2} \ln \left(1+x^2\right)$ $\frac{dy}{dx} = \frac{1}{x} + \frac{1}{2} \left(\frac{1}{1+x^2}\right) (2x)$ $\frac{dy}{dx} = \frac{1}{x} + \frac{x}{1+x^2}$ $\frac{dy}{dx} = \frac{1+x^2+x^2}{x(1+x^2)}$ $\frac{dy}{dx} = \frac{1+2x^2}{x(1+x^2)}$	K1 J1 K1 J1	
Total		9 marks	

4

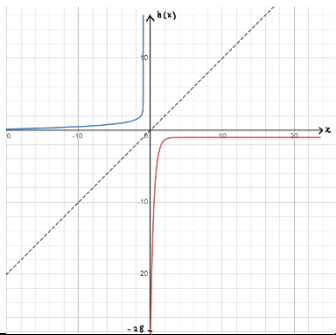
NO.	SOLUTION	MARKS	REMARKS
	$f(-1) = 2(-1) - 6(-1)^{\frac{1}{3}} = 4$ $f(0) = 2(0) - 6(0)^{\frac{1}{3}} = 0$ $f(1) = 2(1) - 6(1)^{\frac{1}{3}} = -4$ $\therefore (-1, 4), (0, 0)$ and $(1, -4)$ are the coordinates of the critical points.	K1 J1	
Total		7 marks	
PART B			
1	$z_1 = 3 + 4i$ $z_2 = 5 - 2i$ $z_1 z_2 = (3 + 4i)(5 - 2i) = 23 + 14i$ $\frac{1}{z_3} = \frac{1}{z_1} + \frac{1}{z_1 z_2}$ $\frac{1}{z_3} = \frac{1}{3 + 4i} + \frac{1}{23 + 14i}$ $= \left(\frac{1}{3 + 4i} \right) \left(\frac{3 - 4i}{3 - 4i} \right) + \left(\frac{1}{23 + 14i} \right) \left(\frac{23 - 14i}{23 - 14i} \right)$ $= \left(\frac{3 - 4i}{25} \right) + \left(\frac{23 - 14i}{725} \right)$ $= \frac{110 - 130i}{725}$	K1 K1 J1	

NO.	SOLUTION	MARKS	REMARKS
	$z_3 = \left(\frac{725}{110-130i} \right) \left(\frac{110+130i}{110+130i} \right)$ $= \frac{79750 + 94250i}{29000}$ $= \frac{11}{4} + \frac{13}{4}i$	K1 J1	
Total		5 marks	
2	(a) $x \ln 3 = 2 - x \ln 2$ $x \ln 3 + x \ln 2 = 2$ $x(\ln 3 + \ln 2) = 2$ $x(\ln 6) = 2$ $x = \frac{2}{\ln 6}$ $\left \frac{2x-1}{x+3} \right \leq 1$ $\frac{2x-1}{x+3} \leq 1 \quad \cap \quad \frac{2x-1}{x+3} \geq -1$	K1 K1 K1 J1 B1	

NO.	SOLUTION	MARKS	REMARKS
	(b) $\frac{2x-1}{x+3} - 1 \leq 0$ $\frac{x-4}{x+3} \leq 0$ $-3 < x \leq 4$	K1	
	$\frac{2x-1}{x+3} + 1 \geq 0$ $\frac{3x+2}{x+3} \geq 0$ $x < -3 \cup x \geq -\frac{3}{2}$	K1K1	
	$\left\{ x : -\frac{3}{4} \leq x \leq 4 \right\}$	J1	
	Total	11 marks	

8

NO.	SOLUTION	MARKS	REMARKS
4	(a)(i) $\left[\frac{e^{2x} + 1}{e^{2x} - 1} \right]^2 - \left[\frac{2e^x}{e^{2x} - 1} \right]^2 = a$ $\frac{e^{4x} + 2e^{2x} + 1 - 4e^{2x}}{(e^{2x} - 1)^2} = a$ $\frac{e^{4x} - 2e^{2x} + 1}{(e^{2x} - 1)^2} = a$ $\frac{(e^{2x} - 1)^2}{(e^{2x} - 1)^2} = a$ $a = 1$ (a) (ii) $ff^{-1}(x) = x$ $\frac{e^{2f^{-1}(x)} + 1}{e^{2f^{-1}(x)} - 1} = x$ $e^{2f^{-1}(x)} + 1 = x(e^{2f^{-1}(x)} - 1)$ $e^{2f^{-1}(x)} - xe^{2f^{-1}(x)} = -1 - x$ $e^{2f^{-1}(x)}(1 - x) = -1 - x$ $e^{2f^{-1}(x)} = \frac{-(1+x)}{(1-x)}$ $f^{-1}(x) = \frac{1}{2} \ln \left \frac{1+x}{1-x} \right$	K1 K1 K1 J1 K1 K1 J1	

NO.	SOLUTION	MARKS	REMARKS
	(b) (i) $hh^{-1}(x) = x$ $-\left(\frac{1}{3}\right)^{2h^{-1}(x)-3} - 1 = x$ $\left(\frac{1}{3}\right)^{2h^{-1}(x)-3} = -x - 1$ $\ln\left(\frac{1}{3}\right)^{2h^{-1}(x)-3} = \ln -x-1 $ $(2h^{-1}(x)-3)\ln\left(\frac{1}{3}\right) = \ln -x-1 $ $h^{-1}(x) = \frac{\ln -x-1 }{2\ln\left(\frac{1}{3}\right)} + \frac{3}{2}$ $h^{-1}(x) = -\frac{\ln -x-1 }{2\ln 3} + \frac{3}{2}$ (b) (ii) 	K1 K1 K1 J1 R1 R1 R1	
Total		14 marks	

NO.	SOLUTION	MARKS	REMARKS
5	(a) $f(c)$ is defined $\lim_{x \rightarrow c} f(x)$ exist $\lim_{x \rightarrow c} f(x) = f(c)$ (b) $\lim_{x \rightarrow 3^-} (k) = \lim_{x \rightarrow 3^+} (x - 2k) = f(3)$ $3 - 2k = k$ $3k = 3$ $k = 1$ (c) $\lim_{x \rightarrow 2^-} \frac{2+x}{4-x^2} = \lim_{x \rightarrow 2^-} \frac{2+x}{(2+x)(2-x)}$ $= \lim_{x \rightarrow 2^-} \frac{1}{2-x}$ $= +\infty$ vertical asymptote $x = 2$	B1 B1 K1 J1 K1 K1 J1	Any properties All properties correct

12

NO.	SOLUTION	MARKS	REMARKS
	(e)	R1 R1 R1	
Total		14 marks	
6	(a) $y = e^{-4x} \ln(2x)$ $u = e^{-4x} \quad v = \ln(2x)$ $u' = -4e^{-4x} \quad v' = \frac{1}{x}$ $\frac{dy}{dx} = (e^{-4x})\left(\frac{1}{x}\right) + (\ln 2x)(-4e^{-4x})$ $\frac{dy}{dx} = (e^{-4x})\left(\frac{1}{x} - 4 \ln 2x\right)$	K1 J1 K1 J1	

NO.	SOLUTION	MARKS	REMARKS
	$(e^{4x}) \frac{dy}{dx} = x^{-1} - 4 \ln 2x$ $(e^{4x}) \left(\frac{d^2y}{dx^2} \right) + \left(\frac{dy}{dx} \right) (4e^{4x}) = -x^{-2} - 4 \left(\frac{1}{2x} \right) (2)$ $(e^{4x}) \left(\frac{d^2y}{dx^2} + 4 \frac{dy}{dx} \right) = -\frac{1}{x^2} - \frac{4}{x}$ $(e^{4x}) \left(\frac{d^2y}{dx^2} + 4 \frac{dy}{dx} \right) = - \left(\frac{x + 4x^2}{x^3} \right)$ $(e^{4x}) \left(\frac{d^2y}{dx^2} + 4 \frac{dy}{dx} \right) = - \frac{x(1 + 4x)}{x^3}$ $(e^{4x}) \left(\frac{d^2y}{dx^2} + 4 \frac{dy}{dx} \right) = - \left(\frac{1 + 4x}{x^2} \right)$ (b) $x = \frac{1}{t^2 - 1} \qquad y = \frac{1 + t^2}{2t}$ $x = (t^2 - 1)^{-1} \qquad u = 1 + t^2 \quad v = 2t$ $\qquad \qquad \qquad u' = 2t \quad v' = 2$	K1 K1 K1 K1 J1	

NO.	SOLUTION	MARKS	REMARKS
	$\frac{dx}{dt} = -1(t^2 - 1)^{-2} (2t) \quad \frac{dy}{dt} = \frac{(2t)(2t) - (1+t^2)(2)}{(2t)^2}$ $\frac{dx}{dt} = \frac{-2t}{(t^2 - 1)^2} \quad \frac{dy}{dt} = \frac{4t^2 - (2 + 2t^2)}{4t^2} = \frac{2t^2 - 2}{4t^2} = \frac{t^2 - 1}{2t^2}$ $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$ $\frac{dy}{dx} = \frac{t^2 - 1}{2t^2} \times \frac{(t^2 - 1)^2}{-2t}$ $\frac{dy}{dx} = \frac{(t^2 - 1)^3}{-4t^3}$ $\frac{dy}{dx} = \frac{(t^2 - 1)^3}{-4t^3} = 0$ $(t^2 - 1)^3 = 0$ $t^2 - 1 = 0$ $t = \pm 1, t > 0 \therefore t = 1$	K1 K1 J1 K1 J1 K1 J1	
Total		16 marks	

NO.	SOLUTION	MARKS	REMARKS
7	(a) $\frac{1}{R} = \frac{1}{P} + \frac{1}{Q}$ $\frac{d}{dt}\left(\frac{1}{R}\right) = \frac{d}{dt}\left(\frac{1}{P}\right) + \frac{d}{dt}\left(\frac{1}{Q}\right)$ $-\frac{1}{R^2} \frac{dR}{dt} = -\frac{1}{P^2} \frac{dP}{dt} - \frac{1}{Q^2} \frac{dQ}{dt}$ $\frac{1}{R^2} \frac{dR}{dt} = \frac{1}{P^2} \frac{dP}{dt} + \frac{1}{Q^2} \frac{dQ}{dt}$ (b) $\frac{1}{R} = \frac{1}{P} + \frac{1}{Q}$ $P = 30, Q = 60$ $\frac{1}{R} = \frac{1}{30} + \frac{1}{60} = \frac{1}{20} \Rightarrow R = 20$ $\frac{1}{R^2} \frac{dR}{dt} = \frac{1}{P^2} \frac{dP}{dt} + \frac{1}{Q^2} \frac{dQ}{dt}$	K1K1 J1 K1J1 K1K1	

NO.	SOLUTION	MARKS	REMARKS
	$\frac{1}{20^2} \frac{dR}{dt} = \frac{1}{30^2} (0.01) + \frac{1}{60^2} (0.03)$ $\frac{1}{20^2} \frac{dR}{dt} = \frac{7}{360000}$ $\frac{dR}{dt} = \frac{7}{900} \text{ ohm s}^{-1}$	K1 J1	
Total		9 marks	